# Banana Leaf Disease Classification from Smartphone Photos using Transfer Learning (EfficientNetB0)

*CBL – Deep Learning | Image Classification Challenge | Agriculture & Food*

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| **Section** | AIML -B | **PS No. & Title** | [PS01] – Banana Leaf Disease Classifier |
| **Date** | 23 October 2026 | **Dataset** | BananaLSD (Mendeley Data) — 937 original field images, 224 × 224 |
| **GitHub** | https://github.com/SwasthikaSelvakumar/banana-leaf-classification/blob/main/BananaLSD_Classifier_.ipynb | | |

## 1. Problem and Objective

Banana growers lose yield when leaf-spot diseases are not caught early. The objective is a CNN that identifies the disease from a smartphone photo of a banana leaf (4 classes: Healthy, Sigatoka, Cordana, Pestalotiopsis). Success is judged by **macro F1 on the held-out test set** (the classes are imbalanced), reported together with test accuracy. The specific challenge is **data leakage**: the Augmented set is generated from the same photos as the Original set and must not reach the test split.

## 2. Dataset and Pre-processing

BananaLSD (Arman et al., Mendeley Data, V1; https://data.mendeley.com/datasets/9tb7k297ff/1) holds 937 original 224×224 smartphone photographs from banana fields in Bangladesh, labelled by a plant pathologist. **Only the Original set was used**; the Augmented set was never loaded. Exact duplicate files were removed by MD5 hash (937 → 898 images; 38 of the 39 duplicates were Healthy leaves, which would otherwise have leaked across splits). A stratified 70/15/15 split (seed 42) gave 628 training, 135 validation and 135 test images; the split was made before any augmentation. Images are passed to EfficientNet in the 0–255 range (its preprocessing is built in). Augmentation is applied to the **training stream only**: flips, rotation (±54°), zoom (±15%), translation (±10%), contrast and brightness jitter. Class imbalance (Sigatoka is the majority class, 71 of 135 test images) was handled with stratified splits, class-weighted loss and macro-averaged metrics.

## 3. Methodology

An ImageNet-pretrained **EfficientNetB0** backbone (4.05 M parameters) feeds a head of global average pooling, BatchNorm, Dropout(0.4), Dense(256, ReLU, L2), Dropout(0.3) and a 4-way softmax (4.38 M parameters in total). Training had two stages: (1) backbone frozen and only the head trained (0.33 M trainable parameters); (2) the last 40 backbone layers unfrozen at a 100× lower learning rate, with BatchNorm layers kept frozen for stability on small data. Early stopping on validation accuracy restored the best weights. Hyperparameters are in Table 1.

**Table 1.** Training configuration.

| Setting | Value | Setting | Value |
|---|---|---|---|
| **Backbone** | EfficientNetB0 (ImageNet weights) | **Input / batch size** | 224×224×3 / 32 |
| **Optimiser** | Adam | **Learning rate (stage 1 / 2)** | 1e-3 / 1e-5 |
| **Loss** | Categorical CE, label smoothing 0.1, class weights | **Epochs run (both stages)** | 28 (19 + 9; max 20 + 40) |
| **Regularisation** | Dropout 0.4 / 0.3, L2 1e-4 | **Early stopping** | val. accuracy, patience 8, best weights restored |
| **Fine-tuned layers** | Last 40 backbone layers (BatchNorm frozen) | **LR schedule** | ×0.3 on val. loss plateau (3 epochs) |

## 4. Results

**Table 2.** Results on the **test set** (n = 135), evaluated once after training.

| Class (test set) | Precision | Recall | F1-score | Support |
|---|---|---|---|---|
| Cordana | 0.96 | 0.96 | 0.96 | 25 |
| Healthy | 1.00 | 1.00 | 1.00 | 13 |
| Pestalotiopsis | 0.96 | 1.00 | 0.98 | 26 |
| Sigatoka | 1.00 | 0.99 | 0.99 | 71 |
| **Macro average** | **0.98** | **0.99** | **0.98** | **135** |

| **EfficientNetB0** — test accuracy **0.99** (133/135) · macro F1 **0.98** · training time 6.2 min (371 s, Colab T4 GPU) |
|---|

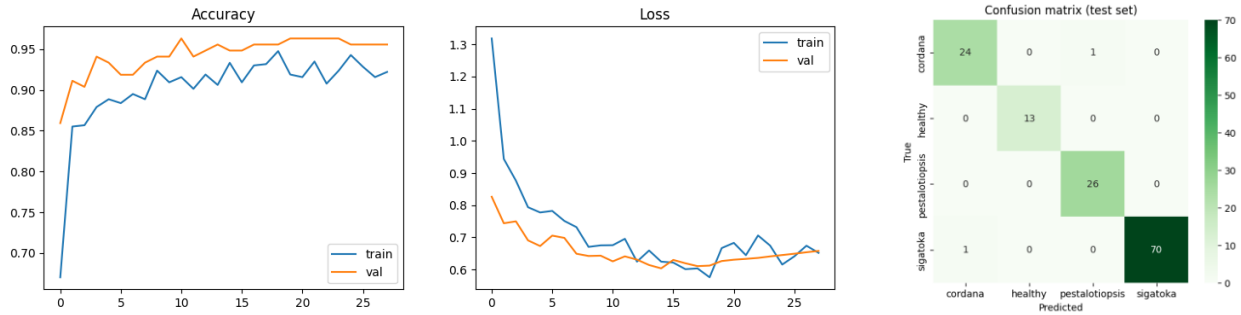



**Fig. 1.** (a) Training and validation accuracy and loss over the 28 epochs of both stages (loss includes label smoothing). (b) Confusion matrix on the test set.

## 5. Analysis and Discussion

Only 2 of 135 test images were misclassified (Fig. 2): a Sigatoka leaf predicted as Cordana (confidence 0.41) and a Cordana leaf predicted as Pestalotiopsis (0.35). Healthy leaves were separated perfectly and no diseased leaf was labelled healthy. Both errors involve a **sparse lesion**: the Sigatoka image shows only two or three tiny brown spots at the leaf edge against leaf litter (early-stage, little visual evidence), and the Cordana image shows one isolated dark lesion with a yellow halo on an over-exposed yellow-green leaf, resembling a Pestalotiopsis spot. Both were low-confidence, as were the two shakiest correct cases (0.44, 0.49: a single small lesion or an unusual colour cast), so a 0.50 threshold would route all four to an expert. **Fit:** validation accuracy (≈0.96) exceeds training accuracy (≈0.92) because training metrics are measured under augmentation, dropout and label smoothing; the loss curves converge, so there is no over- or under-fitting. Validation loss bottomed out at epoch 15 (0.60) and drifted to 0.66 while accuracy stayed flat (mild over-confidence), so early stopping restored the best checkpoint; stage-2 fine-tuning added no validation gain. **Leakage:** only Original images were used, duplicates were removed, the split precedes augmentation, and the test set was evaluated once. **Caveat:** with 135 test images one error is 0.74 points (95% Wilson interval for accuracy: 0.95–1.00), and with no plant IDs, near-duplicate shots of one plant may span splits, so 0.99 may be optimistic.

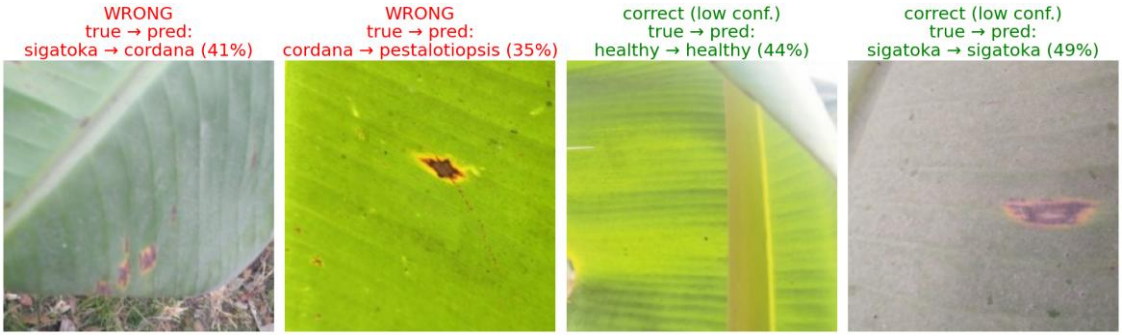



**Fig. 2.** Four test images with true and predicted label and confidence: the two errors (red) and the two lowest-confidence correct predictions (green); all four are below 0.50.

## 6. Real-world Relevance, Limitations and Ethics

A compact model (4.4 M parameters) could be converted to TensorFlow Lite for offline, in-field scouting by smallholders and extension workers. It is not deployable yet: all photos come from one region, one month (June 2021) and three phones; there is no “other/unknown” class, so another disease or nutrient deficiency is forced into one of four labels; and severity is not estimated. Wrong advice can cause missed treatment or unnecessary fungicide use, so predictions should be confidence-gated and confirmed by an expert; farm photos can carry GPS metadata, which should be stripped to protect privacy.

## 7. Conclusion and Future Work

Two-stage fine-tuning of EfficientNetB0 on the leakage-free Original split reached 0.99 test accuracy and 0.98 macro F1 on 135 unseen images. Next, I would test on photos from a different farm and season (domain shift) and add Grad-CAM to check that the model attends to lesions rather than background.

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**Dataset:** Arman, S. E., Bhuiyan, M. A. B., Abdullah, H. M., et al. (2023). Banana Leaf Spot Diseases (BananaLSD) Dataset for Classification of Banana Leaf Diseases Using Machine Learning. Mendeley Data, V1, doi:10.17632/9tb7k297ff.1; and “BananaLSD: A banana leaf images dataset…”, Data in Brief (2023).